TeleGlove: Bidirectional-Glove for Real-Time Hand Motion Tracking and Haptic Feedback

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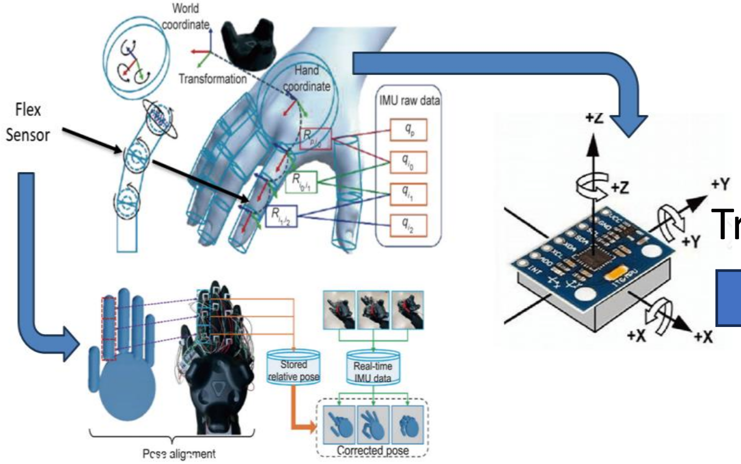
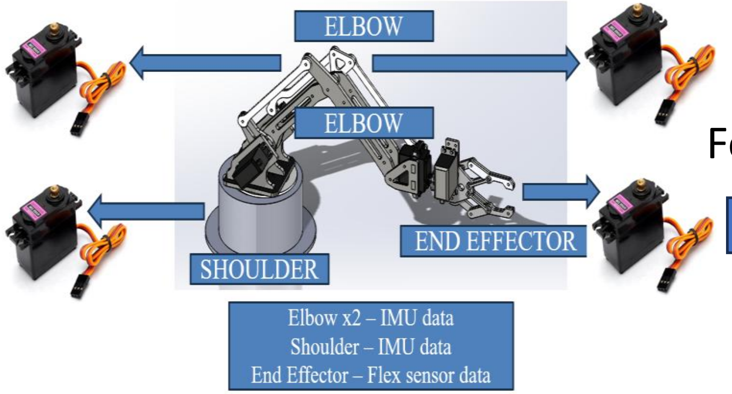
Developed a wearable haptic glove integrating force-sensitive resistors at the fingertips, flex sensors for finger joint movement, and an IMU mounted on the dorsal wrist to capture hand position and finger-bending.

Sensor data were fused to map position and motion within a real-time virtual environment implemented in Unity, enabling visualization and interaction such as grasping, squeezing, and throwing of virtual objects.

The glove used vibrotactile motors at the outer fingertips to form a closed-loop system by providing a force-proportional haptic feedback based on measured fingertip contact forces and finger flexion.



Motion Translation



Haptic Feedback

Sensor Integration: FSR, IMU, Flex sensors

3 DoF Robotic Arm Actuator

Haptic Feedback

Achieved 65% motion translation and feedback efficiency from real-time hand gestures to system response.